



Vikash Kumar
Chemical Engineering
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B.Tech.
Gender: Male
DOB: 10/07/2003

Examination	University	Institute	Year	CPI / %
Graduation	IIT Bombay	IIT Bombay	2026	

SCHOLASTIC ACHIEVEMENTS

- Ranked among top **3%** in JEE Advanced among more than **0.15 million** candidates across all over India ('22)
- Secured a percentile of **98.8** in JEE Main attempted by more than **1 million** students across all over India ('22)
- Awarded **Certificate of Merit** for placing in the **top 9** for the **Worley IICHE-MRC Best Design Project Award** ('25)

RESEARCH EXPERIENCE

Automation For Road Development | *RnD Project* | *Prof. Milind A. Sohoni* (Jan '24 - Apr '25)

- Designed, Developed and published a **Flutter** based app on **Google Play Store** to assess road pavement conditions
- Managed offline data storage with **SQLite** to save sensor readings when internet is unavailable and sync them later
- Integrated **Google Maps API** to show road routes and attributes, and **Google Drive API** for data backup and sync
- Developed a **PDF report generation** feature to help teams plan and present road assessment data to management

KEY PROJECTS

Facial Expression Recognition | *Season of Code* | *WnCC* (Jun '24 - Jul '24)

- Developed a **CNN** based facial expression model to recognize **7** emotions using **PyTorch** on the FER-2013 Dataset
- Effectively leveraged transfer learning to fine-tune **VGG16** and **ResNet50** achieving top accuracy of **66%** with ResNet
- Integrated **OpenCV** for real time video processing to detect and classify facial expressions from live camera feed

Maximum Matching in Graph | *Course Project* | *Prof. Ashutosh Mahajan* (Jan '25 - Apr '25)

- Implemented **Edmonds' Blossom Algorithm** in **C++** for maximum matching in general graphs to handle odd cycles
- Designed an **OOP** based graph and matching interface with **LCA** detection for modular, maintainable implementation
- Evaluated performance on synthetic graph from **5k-500k** vertices by plotting runtimes for analyzing time complexity

Shape File Parser | *Self Project* | *GIS Application Development* (Dec '24 - Jan '25)

- Developed a **Flutter** application to parse and visualize **ESRI Shapefiles** for easy exploration of geospatial datasets
- Implemented **parsing utils** in **Dart** to handle complex geometries including polygon with internal rings and polylines
- Implemented a feature inspection tool for **metadata analysis**, improving user understanding of geospatial datasets

Railway Planner | *Self Project* | *Data Structures & Algorithms* (Jun '25 - Jul '25)

- Built a railway planning system with an **OOP architecture**, integrating **graphs, tries, BSTs, heaps, and hash maps**
- Implemented multi-criteria route planning with Dijkstra's algorithm and **BFS** for distance, time, and cost computation
- Built a **trie-powered search engine** with auto-completion and with case-insensitive matching for faster station lookup

TECHNICAL SKILLS

- Languages** : C | C++ | Dart | Python | MATLAB | SQL | $\LaTeX$
- Libraries** : NumPy | Pandas | SciPy | OpenCV | Matplotlib | Scikit-Learn | PyTorch | TensorFlow
- Tools & Frameworks** : Flutter | Android Studio | Firebase | pgAdmin | PostgreSQL | Git | GitHub | gdb | Typst
- Software & Platforms** : DWSIM | ASPEN | Fusion 360 | QGIS | AutoCAD | COMSOL

EXTRA CURRICULAR ACTIVITIES

- Mentored **10+** students in building an app using Flutter framework that aims to connect alumni with current students
- Completed **80+** hours of community service with **NSS IIT Bombay**, tutoring Math and Chemistry to **6** students
- Built foundational knowledge in **AI** technologies through hands-on experience in the **Microsoft Copilot workshop**
- Participated in article writing competition **Abhivyakti 2023** organized by Sailesh J. Mehta School Of Management